

Operations Research 2017/2018

Exercises Sheet nº3 Duality and dual Simplex method

1. Formulate the dual problem for each of the following linear programming problems:

a) Maximize $z = 3x_1 + 2x_2$

subject to

$$x_1 \leq 4$$

$$x_1 + 3x_2 \leq 15$$

$$2x_1 + x_2 \leq 10$$

$$x_1, x_2 \geq 0$$

b) Minimize $z = 5x_1 + 4x_2 - x_3 + 3x_4$

subject to

$$3x_1 + 2x_2 - 3x_3 + x_4 \leq 24$$

$$3x_1 + 3x_2 + x_3 + 3x_4 \geq 36$$

$$x_1, x_2, x_3, x_4 \geq 0$$

c) Maximize $z = 10x_1 + 20x_2$

subject to

$$-x_1 + 2x_2 \leq 15$$

$$x_1 + x_2 \leq 12$$

$$5x_1 + 3x_2 \leq 45$$

$$x_1, x_2 \geq 0$$

d) Maximize $z = 2x_1 - 2x_2 + 3x_3$

subject to

$$-x_1 + x_2 + x_3 \leq 4$$

$$2x_1 - x_2 + x_3 \geq 2$$

$$x_1 + x_2 + 3x_3 \leq 12$$

$$x_1, x_2, x_3 \geq 0$$

e) Minimize $z = 12x_1 + 10x_2$

subject to

$$4x_1 + 5x_2 \geq 2$$

$$3x_1 + 2x_2 = 3$$

$$-x_1 + x_2 \leq 1$$

$$x_1, x_2 \geq 0$$

f) Minimize $z = x_1 - 4x_2$

subject to

$$-3x_1 + x_2 \leq 6$$

$$x_1 + 2x_2 = 4$$

$$x_1, x_2 \geq 0$$

g) Maximize $z = -x_1 + 2x_2 + x_3$

subject to

$$3x_2 + x_3 \leq 120$$

$$x_1 - x_2 - 4x_3 \geq 80$$

$$-3x_1 + x_2 + 2x_3 \leq 100$$

h) Minimize $z = 3x_1 + 2x_2 + 4x_3$

subject to

$$2x_1 + x_2 + 3x_3 = 60$$

$$3x_1 + 3x_2 + 5x_3 \geq 120$$

$$x_1, x_2, x_3 \geq 0$$

2. Consider the following LP problem:

Maximize $z = 6x_1 + 8x_2$

subject to

$$5x_1 + 2x_2 \leq 20$$

$$x_1 + 2x_2 \leq 10$$

$$x_1, x_2 \geq 0$$

- Formulate the dual problem for this primal problem.
- Solve the primal and dual problems by the graphical method. Identify the feasible and non-feasible EPs (extreme points) for both problems. Calculate the value of the objective functions for all of these points.
- Solve the dual problem by the Simplex method (using the “Big M” technique). Confirm if the final results are the same as the ones obtained in b).

3. Consider the following LP problem:

Maximize $z = x_1 + 2x_2$

subject to

$$x_1 + 3x_2 \leq 8$$

$$x_1 + x_2 \leq 4$$

$$x_1, x_2 \geq 0$$

- Formulate the dual problem for this primal problem.
- Solve the primal and dual problems by the graphical method. Identify the feasible and non-feasible EPs (extreme points) for both problems. Calculate the value of the objective functions for all of these points.
- Solve the dual problem by the Simplex method (using the “Big M” technique). Confirm if the final results are the same as the ones obtained in b).

4. Consider the following LP problem:

Minimize $z = x_1 + x_2$

subject to

$$x_1 + x_2 \leq 8$$

$$x_2 \geq 3$$

$$-x_1 + x_2 \leq 2$$

$$x_1, x_2 \geq 0$$

- Solve the problem using the graphical method;

- b) Solve it again using the dual Simplex method;
- c) Formulate the dual problem associated with it;
- d) Without solving the dual problem, determine its optimal solution. Justify the obtained values.

5. Consider the following LP problem:

$$\text{Minimize } z = 5x_1 + 2x_2 + 4x_3$$

subject to

$$3x_1 + x_2 + 2x_3 \geq 4$$

$$6x_1 + 3x_2 + 5x_3 \geq 10$$

$$x_1, x_2, x_3 \geq 0$$

- a) Solve it using the dual Simplex method;
- b) Formulate the dual problem associated with it;
- c) Without solving the dual problem, determine its optimal solution. Justify the obtained values.

6. Consider the following LP problem:

$$\text{Maximizar } z = 3x_1 + 2x_2$$

sujeito a

$$3x_1 + x_2 \leq 12$$

$$x_1 + x_2 \leq 6$$

$$5x_1 + 3x_2 \leq 27$$

$$x_1, x_2 \geq 0$$

- a) Formulate the dual problem associated with it;
- b) Solve the dual problem using the dual Simplex method;
- c) Without solving the primal problem, determine its optimal solution. Justify the obtained values.

7. Consider the following LP problem:Minimize $z = x_1 + 2x_2 + 4x_3$

subject to

$$x_1 + 3x_2 \leq 5$$

$$x_1 + 3x_3 \geq 4$$

$$x_2 + x_3 \leq 9$$

$$x_1, x_2, x_3 \geq 0$$

- a) Solve it using the dual Simplex method;
- b) Formulate the dual problem associated with it;
- c) Without solving the dual problem, determine its optimal solution. Justify the obtained values.

8. Consider the following LP problem:Minimize $z = 4x_1 + x_2$

subject to

$$3x_1 - x_2 \leq 5$$

$$x_1 + 4x_2 \geq 3$$

$$2x_1 + x_2 \leq 4$$

$$x_1 \geq 0, x_2 \geq 0$$

- a) Solve the problem using the graphical method;
- b) Solve it again using the dual Simplex method;
- c) Formulate the dual problem associated with it;
- d) Without solving the dual problem, determine its optimal solution. Justify the obtained values.

9. Consider the following LP problem:

Minimize $z = 2x_1 + 3x_2 + x_3$

subject to

$$2x_1 + x_2 + 3x_3 \leq 8$$

$$x_1 + 3x_3 \geq 6$$

$$3x_1 + 2x_2 \leq 2$$

$$x_1, x_2, x_3 \geq 0$$

- a) Solve it using the dual Simplex method;
- b) Formulate the dual problem associated with it;
- c) Without solving the dual problem, determine its optimal solution. Justify the obtained values.

10. Consider the following LP problem:

Minimize $z = 2x_1 + x_2 + 3x_3 + x_4$

subject to

$$2x_1 + 3x_2 + x_3 + x_4 \leq 6$$

$$x_1 + x_3 \geq 2$$

$$x_1 + 2x_2 + x_3 + 2x_4 \leq 4$$

$$x_1, x_2, x_3, x_4 \geq 0$$

- a) Solve it using the dual Simplex method;
- b) Formulate the dual problem associated with it;
- c) Without solving the dual problem, determine its optimal solution. Justify the obtained values.